Проверочная – 1

Time allowed: 2 hours

1. Let A and B be $n \times n$ matrices with complex entries for which

$$A^2 + B^2 = 2(AB - BA).$$

Prove that $A^2 + B^2$ is a nilpotent matrix.

2. Let (a_n) be a recursive sequence given by

$$a_0 = 1, \qquad a_{n+1} = \frac{2a_n}{2 + \sqrt{4 - a_n^2}}.$$

Prove that $(2^n a_n)$ is increasing and find its limit.